ZCAS University

CIT 2341 - CRIME MAPPING

FINAL EXAMINATION

27TH MAY 2026

09:00 - 12:00 HOURS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. **Section A:** This section has only one question, which is compulsory and carries **40 Marks**
2. **Sections B**: Answer any **THREE** (3) questions from this section. Each question in this section carries 20 Marks
3. The total Marks for this exam are 100.
4. This exam paper consists of **FIVE** (5) Questions
5. Do not turn this page until you are told to do so.

QUESTION 1

SECTION A: COMPULSORY

Case Study: Cybercrime Data Collection and Analysis at Lusaka Digital Forensics Unit

The Lusaka Digital Forensics Unit (LDFU) was assigned to investigate a rapid increase in online banking fraud and mobile-money scams reported across Lusaka Province. The unit discovered that victims were reporting incidents through multiple channels, including police stations, mobile network operators, and online complaint portals. However, investigators faced challenges because some reports were incomplete, duplicated, or inconsistent. To improve investigations, the unit deployed a research team composed of cybersecurity analysts, systems analysts, and crime-mapping specialists. The team conducted interviews with fraud victims, distributed online questionnaires to affected citizens, reviewed police incident records, observed operations at mobile-money booths, and collected samples of transaction logs from financial institutions. The team also used both quantitative and qualitative research methods to understand the scale of cybercrime and the experiences of victims. Quantitative methods were used to analyze the number of fraud cases, financial losses, and geographic distribution of attacks, while qualitative methods were used to understand victim experiences, emotional impacts, and public perceptions of cybersecurity awareness. During the investigation, the analysts discovered that some junior officers lacked technical skills in scripting and data analysis, while others struggled with communication and interpersonal skills during victim interviews.

a) Identify FOUR data collection methods used in the case study and explain how each method contributed to the cybercrime investigation. **(8 Marks)**

b) Differentiate between structured interviews and unstructured interviews. State ONE advantage and ONE disadvantage of each approach in the context of cybercrime investigations. **(6 Marks)**

c) Discuss the difference between quantitative research and qualitative research as applied in the case study. Provide TWO examples of data collected under each method. **(8 Marks)**

d) Describe FOUR essential skills a cybersecurity officer or systems analyst should possess when conducting cybercrime investigations and data analysis. **(8 Marks)**

e) Explain THREE advantages and TWO disadvantages of qualitative research in understanding victim experiences related to cybercrime. **(5 Marks)**

f) Propose recommendations that the Lusaka Digital Forensics Unit can implement to improve future cybercrime data collection, analysis, and reporting processes. **(5 Marks)**

[40 MARKS]

QUESTION 2

a) There are many different statistical techniques that are used, depending on the specific needs of the analysis. The common ones are descriptive, inference and predictive statistics techniques. Explain each of the techniques described above. **(9 Marks)**

b) What is the difference between Population and sample size in statistics? Draw a diagram to show the difference. **(6 Marks)**

c) What is the significance of statistical analysis and the P-value? **(5 Marks)**

[MARKS 20]

QUESTION 3

a) Discuss the differences between global and local measures of spatial autocorrelation. Include suitable examples. **(4 Marks)**

b) Explain the concept of Moran's I and describe how it is used to identify clustered, random, and dispersed spatial patterns. **(4 Marks)**

c) Using appropriate diagrams, explain the following spatial patterns according to Moran's I:

i. a) Clustered pattern
ii. b) Random pattern
iii. c) Dispersed pattern

(6 Marks)

d) A researcher obtains a Moran's I value of 0.72 with a Z-score of 2.8. Interpret the result. **(2 Marks)**

e) A spatial analyst obtains a Moran's I value of -0.65. Explain what this indicates about the geographic distribution of the data. **(2 Marks)**

f) Explain why Moran's I is considered a global measure and not a local measure of spatial autocorrelation. **(2 Marks)**

QUESTION 4

a) List and describe the four types of spatial data. **(8 Marks)**

b) In spatial data, there are two models of data representation: raster and vector. Draw a diagram and explain the difference between each of them to demonstrate your understanding. **(8 Marks)**

c) The figures below show the types of data in GIS. Name each of the figures below to demonstrate your understanding of GIS data types. **(4 Marks)**

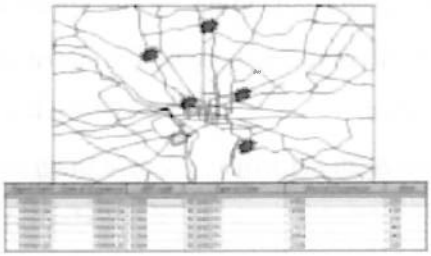


Figure 1

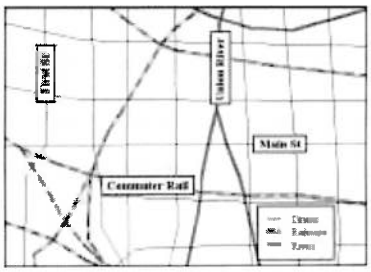


Figure 3

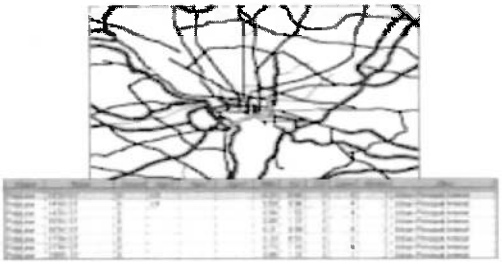


Figure 2

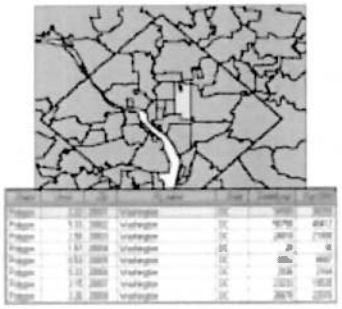


Figure 4

[MARKS 20]

QUESTION 5

MULTIPLE CHOICE QUESTIONS

1. Which of the following best describes a global measure of spatial autocorrelation?
   A. A value calculated for each observation unit
   B. A measure that applies only to polygons
   C. A single value representing the entire geographic dataset
   D. A method used only for binary variables
2. Which spatial autocorrelation statistic is most commonly used for continuous variables?
   A. Join Count Statistic
   B. Moran's I
   C. Chi-square Test
   D. Geary's Ratio
3. Moran's I values generally range between:
   A. 0 and 1

B. –1 and +1

C. –10 and +10

D. 1 and 100

4. A Moran's I value close to +1 indicates:

A. Random spatial pattern

B. Negative spatial autocorrelation

C. Positive spatial autocorrelation

D. No clustering

5. Which of the following variables is suitable for Moran's I analysis?

A. Binary variable only

B. Continuous variable

C. Nominal variable only

D. Text data

6. A Moran's I value near –1 indicates:

A. Clustered pattern

B. Random pattern

C. Dispersed pattern

D. Hot spot pattern

7. Which statistic is mainly used for binary spatial data?

A. Moran's I

B. Geary's C

C. Join Count Statistic

D. Regression Analysis

8. In Moran Scatterplots, the slope of the regression line represents:

A. Correlation coefficient

B. Standard deviation

C. Moran's I

D. Mean spatial lag

9. Hot spots refer to:

A. Areas where low values cluster together

B. Areas with random distribution

C. Areas where high values cluster together

D. Areas with no spatial autocorrelation

10. The null hypothesis in Moran's I significance testing states that:

A. Spatial autocorrelation exists

B. Data are clustered

C. No spatial autocorrelation exists

D. All variables are positively correlated

[MARKS 20]

END